Assignment: Version Control with Git

EGR 498

Complete this assignment **after the guest lecture**. Add your GitHub username and correct your hometown in your team's roster. You will save a file, record a commit, and share it with your team. Use a Windows or Mac computer. No Raspberry Pi or Python is needed.

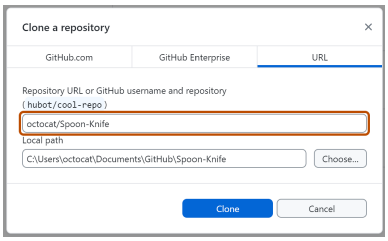
Your team's repository: Use the link for your team below.
Blue: `github.com/mmurphy39-hash/Senior-Design-Blue`
Gold: `github.com/BraxtonAlexander0/Senior-Design-Gold-Test`

1 Install the Software and Get Access

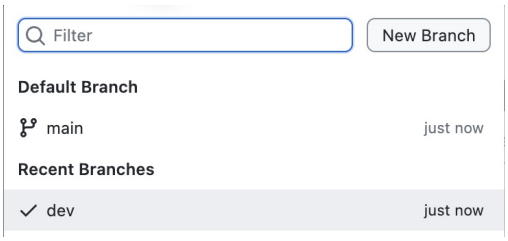
1. **Install GitHub Desktop on your computer.** Download it from `desktop.github.com` and run the installer.
2. **Create an account on GitHub.** Go to `github.com/signup` and complete the setup, including verifying your email. Use your existing account if you have one.
3. **Email your software coordinator** your name, Blue or Gold team, role number, and **GitHub username**. The coordinator needs your username to give you access.
4. **Accept the repository invitation.** Open GitHub's invitation email, sign into the account from step 2, and accept. Wait until you have access before continuing.
5. **Open Desktop and sign in.** Choose **Sign in to GitHub.com** and follow the browser prompts using that same account. If asked to configure Git, enter your name and the email associated with your GitHub account.

2 Download the Team Files and Select dev

6. **Clone your team's repository.** Click **File** → **Clone Repository** → **URL**. Paste your team's link from the top of this prompt. **Local Path** is the folder where the files will go. Use the suggested path or choose a folder. Click **Clone**. If already cloned, select it under **Current Repository**.
7. **Select the existing dev branch.** Click **Current Branch**, then **dev** in the list. Confirm that the toolbar says **dev**. This is called *checking out* a branch: it selects that branch's files in your local folder. Do not click **New Branch**.
8. **Get the latest changes.** Click **Fetch origin**. If **Pull origin** appears, click it. Fetch checks GitHub for new commits. Pull brings them into your local branch. If Pull does not appear, you are up to date.



(a) Paste your team's link in the URL field (GitHub Docs).



(b) Click the existing `dev` branch (practice repository).

Lecture materials: `github.com/tstoffel1/egr498-git-tutorial`. For additional help, see the **longer assignment guide** there under `team-assignment`.

3 Edit Your Row and Commit the Change

9. **Open and edit the roster.** Choose **Repository** → **Show in Explorer** (Windows) or **Show in Finder** (Mac). Open `team-roster.txt` in Notepad or TextEdit. In TextEdit, use **Format** → **Make Plain Text** if needed. A monospaced font with word wrapping off makes the columns easier to follow.

   Find your **role number**. Fill in your GitHub username and replace the example hometown with your own. Change **only your row**, keeping spaces between columns (no Tab). **Save** with the existing filename.

10. **Check the changes in Desktop.** Click **Changes**, then `team-roster.txt`. Red shows the old text. Green shows the new text. Check that only your row changed. Correct any unintended edits in your editor and save again.

11. **Commit your change.** Check the box beside `team-roster.txt`. In the **Summary** box, type **Add username and hometown for role 7**, using your own role number. Click the blue commit button. Its label must end with **to dev**. If it says **to main**, stop and ask your coordinator.

Example: A student assigned role 7 fills in the two fields below. Use your own username and hometown.

	Role	GitHub username	Hometown
Before	7		Fresno, CA
After	7	`example-student`	Murray, KY

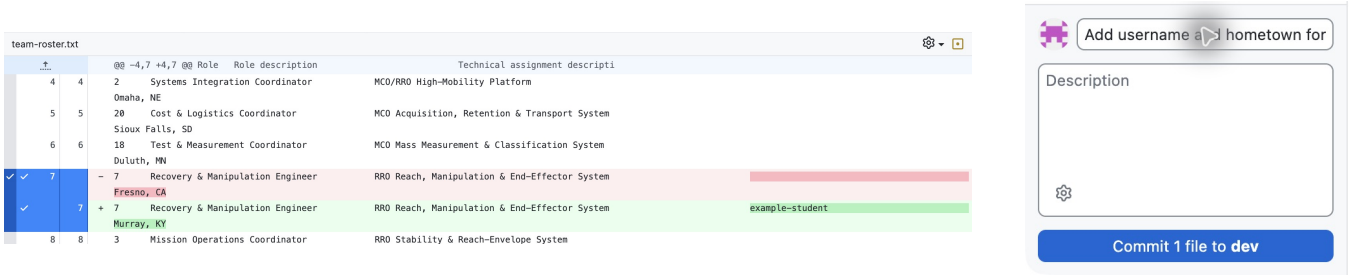


Figure 2: Check the red/green changes, enter a summary, and commit to `dev`.

Save changes the file. Commit records a revision on your computer. Push shares it on GitHub.

4 Share Your Commit and Check the Result

12. **Push your commit.** Click **Fetch origin** again and **Pull origin** if offered, then **Push origin**. This receives classmates' changes before sending yours. If GitHub reports newer changes, repeat Fetch, Pull, and Push.

13. **Verify your update online.** Choose **Repository** → **View on GitHub**. On the website, select **dev** in the branch dropdown above the file list, then open `team-roster.txt`. Confirm your username and hometown appear in your row. **The assignment is complete when your correct entry is visible on GitHub.**

Your coordinator will review `dev` and merge it into `main`. You do not need a new branch, fork, or pull request for this assignment.

If you get stuck: If `dev` is missing, Fetch and check again. For a conflict, missing invitation, or permission/fork message, contact your coordinator with the step and message shown. Keep your saved file and commit. Do not overwrite the roster or delete someone else's entry.